UNIVERSITY OF BRASÍLIA – UnB
GRADUATE PROGRAM IN APPLIED COMPUTING – PPCA
COURSE: IT PROJECT MANAGEMENT (GPTI)
PROFESSORS: Dr. JOÃO MELLO DA SILVA and Dr. EDISON ISHIKAWA

SIPA Privacy 2 Group: Filipe Alves de Sousa, Emanuel Brandão de Galvão Correia, Ruben Isaac Dias da Silva and Carlos Eduardo da Silva Papa

PMBOK 8: PRINCIPLES, FOCUS AREAS, PERFORMANCE DOMAINS AND TAILORING

BRASÍLIA – DF
2026

Abstract

This report synthesizes *The Standard for Project Management* and *A Guide to the Project Management Body of Knowledge*, both from the eighth edition of the PMBOK. Its aim is to explain how six principles guide decisions, five Focus Areas organize activities, and seven performance domains group 40 processes. The reading emphasizes tailoring: choosing the development approach, processes, and forms of engagement that make sense for the organization and the project, and reviewing those choices as results come in. The text covers the value delivery system, life cycles and the choice of approach, the five actions on processes, the four steps of the tailoring process, and the diagnostics in the tailoring chapter. A table relates each of the 40 processes to its domain, Focus Area, and an analytical grouping; a 7×5 map shows the distribution of the processes; a 6×7 matrix summarizes the connections between principles and domains that the *Standard* describes. Four concentrations — Consolidation, Birth, Means, and Who — are used as an **analytical grouping adopted in the GPTI/PPCA course**, without attributing them to PMI. Recent studies on hybrid approaches and an international project management standard situate the synthesis. A final table shows where PMBOK 8 already touches personal data; no new process is proposed here.

Keywords: PMBOK 8; project management; principles; Focus Areas; performance domains; processes; tailoring; hybrid approaches.

Resumo

Este relatório sintetiza *The Standard for Project Management* e *A Guide to the Project Management Body of Knowledge*, ambos da oitava edição do PMBOK. O objetivo é explicar como seis princípios orientam decisões, cinco áreas de foco organizam atividades e sete domínios de desempenho reúnem 40 processos. A leitura enfatiza a adaptação sob medida (*tailoring*): escolher abordagem de desenvolvimento, processos e formas de engajamento que façam sentido para a organização e o projeto, revendo as escolhas conforme os resultados. O texto cobre o sistema de entrega de valor, os ciclos de vida e a escolha da abordagem, as cinco ações sobre processos, os quatro passos do processo de adaptação e os diagnósticos do capítulo de adaptação. Uma tabela relaciona cada um dos 40 processos ao domínio, à área de foco e a um agrupamento analítico; um mapa 7×5 mostra a distribuição dos processos; uma matriz 6×7 resume as conexões entre princípios e domínios que o *Standard* descreve. Quatro concentrações — Consolidação, Nascimento, Meios e Quem — são usadas como **agrupamento analítico adotado na disciplina GPTI/PPCA**, sem atribuí-las ao PMI. Estudos recentes sobre abordagens híbridas e uma norma internacional de gestão de projetos situam a síntese. Um quadro final mostra onde o PMBOK 8 já toca dados pessoais; não se propõe aqui um novo processo.

Palavras-chave: PMBOK 8; gerenciamento de projetos; princípios; áreas de foco; domínios de desempenho; processos; adaptação sob medida; abordagens híbridas.

Sumário

1 PURPOSE, SOURCE, AND METHOD . . . 8
2 STRUCTURE OF THE EDITION AND WHAT CHANGED . . . 9
3 VALUE DELIVERY SYSTEM, ENVIRONMENT, AND FUNCTIONS 10
4 THE SIX PRINCIPLES AND THE MINDSET OF THE *STANDARD* 11
5 LIFE CYCLES, DEVELOPMENT APPROACHES, AND CADENCE 12
5.1 Predictive, adaptive, and hybrid . . . 13
5.2 How to choose and at what cadence to deliver . . . 13
6 FIVE FOCUS AREAS AND THE DISTRIBUTION OF THE PROCESSES . . . 14
7 THE SEVEN PERFORMANCE DOMAINS OF THE *GUIDE* . . . 16
7.1 Auditable inventory of the 40 processes . . . 17
7.2 Tailoring considerations by domain . . . 19
8 HOW THE PRINCIPLES CUT ACROSS THE DOMAINS . . . 19
9 TAILORING . . . 20
9.1 Why tailor . . . 21
9.2 What to tailor . . . 21
9.3 The tailoring process in four steps . . . 22
9.4 Tailoring the domains and running diagnostics . . . 23
10 FOUR CONCENTRATIONS AS AN ANALYTICAL GROUPING . . . 24
11 WHERE PMBOK 8 ALREADY TOUCHES PERSONAL DATA AND PRIVACY . . . 25
12 CONCLUSION . . . 26
12.1 Use of artificial intelligence . . . 26
13 FINAL REPORT OF THE WORK . . . 27

Lista de ilustrações

Figura 1 – Structure of the two works and analytical grouping 9
Figura 2 – 7 × 5 map of the 40 processes 15
Figura 3 – Matrix of principle relationships 20
Figura 4 – Tailoring cycle 22
Figura 5 – Grouping of the seven domains into four concentrations 25

Lista de tabelas

Tabela 1 – Structure of PMBOK 8: Standard and Guide 9
Tabela 2 – Mindset dimensions and principles 11
Tabela 3 – Practical questions and consequences of each principle 12
Tabela 4 – Development approaches: predictive, adaptive and hybrid 13
Tabela 5 – Factors for choosing the development approach 14
Tabela 6 – The five Focus Areas 14
Tabela 7 – Process count by Focus Area 15
Tabela 8 – Inventory of the 40 PMBOK 8 processes 17
Tabela 9 – Characteristic tailoring consideration of each domain 19
Tabela 10 – Hypothetical examples of the five operations on processes 22
Tabela 11 – Common situations and tailoring suggestions 23
Tabela 12 – Analytical concentrations and process count 24
Tabela 13 – PMBOK 8 points that touch personal data 25
Tabela 14 – Final report of the work 27

LIST OF ABBREVIATIONS AND ACRONYMS

1AHV	Adopt a Holistic View
2FOV	Focus on Value
3QPD	Embed Quality Into Processes and Deliverables
4BAL	Be an Accountable Leader
5SPA	Integrate Sustainability Within All Project Areas
6BEC	Build an Empowered Culture
A, S, I, P	Themes under analysis outside PMI: business analysis, sustainability, innovation and privacy
CL	Closing Focus Area
EX	Executing Focus Area
FI	Finance domain
GDPR	General Data Protection Regulation
GO	Governance domain
IA	Artificial intelligence (AI)
IN	Initiating Focus Area
ISO	International Organization for Standardization
MC	Monitoring and Controlling Focus Area
PL	Planning Focus Area
PMBOK	Project Management Body of Knowledge
PMI	Project Management Institute
PMO	Project Management Office
RE	Resources domain
RI	Risk domain
SH	Schedule domain
SIPA	Sustainability, Innovation, Privacy and Business analysis with risk-adjusted decisions
SO	Scope domain
SOX	Sarbanes-Oxley Act
ST	Stakeholders domain
TIC	Information and communication technology (ICT)

Nature of the work. SIPA group: Privacy 2
Members: Filipe Alves de Sousa, Emanuel Brandão de Galvão Correia, Ruben Isaac Dias da Silva and Carlos Eduardo da Silva Papa
Professors: Dr. João Mello da Silva and Dr. Edison Ishikawa
Nature: technical synthesis of PMBOK 8, with an analytical grouping of the domains presented separately from PMI's official structure.

1 PURPOSE, SOURCE, AND METHOD

PMBOK 8 brings together a *Standard*, which presents foundations for project management, and a *Guide*, which organizes performance domains, processes, inputs, outputs, tools and techniques, and tailoring criteria. The purpose of this report is to offer a synthesis of the two works that makes it possible to understand **how the edition is organized and how it can be tailored to a specific project**, with an emphasis on *tailoring*.

Source. The primary source is the complete edition published by PMI in 2025 [5]. It is a **restricted-access document**: the complete edition is obtained through purchase or PMI membership. For this reason, the text cites sections, tables, and figures and avoids reproducing long passages; the descriptions are working paraphrases. The *Guide* indicates, in its introduction, four main sections: performance domains (section 2), tailoring (3), inputs and outputs (4), and tools and techniques (5) [5, Guide §1.1].

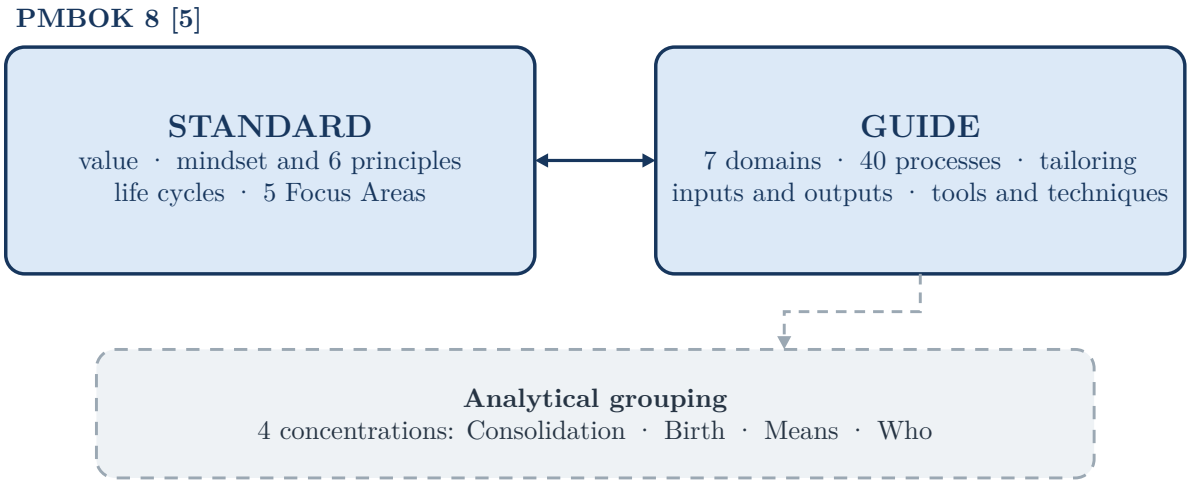
Method. For each element, we asked **what decision it guides**, **where it appears**, and **what result is expected**. We identified the names and the count of the processes in sections 2.1 to 2.7 of the *Guide*, in Table 2-1, and in the overview figures of each domain. The six principles and their connections with the domains were read in chapter 3 of the *Standard*, the life cycles and Focus Areas in chapter 4 of the *Standard*, and *tailoring* in chapter 3 of the *Guide*. Official terms are given in English; the Portuguese renderings that appear next to the official process names in section 7.1 are working translations and are not translations authorized by PMI.

Complementary literature. To situate the synthesis, we used three recent studies on hybrid approaches and tailoring [4, 6, 7], a review on digital twins in construction projects [3], and the ISO 21502:2020 standard, with guidance on project management [2]. These works do not replace PMBOK 8; they serve to compare PMI's reading with what the literature observes in practice.

Limits of the method. The list of the 40 processes is complete to allow the count to be audited, but the report does not reproduce the book. PMBOK 8 states that the inputs and outputs in chapter 4 of the *Guide* are not intended to be comprehensive or mandatory in every project: they are common selections among many options [5, Guide §1.1]. The number of processes presented is therefore not equivalent to the number of

processes to be performed in a project.

Figura 1 – Structure of the two works and analytical grouping



Source: Prepared by the authors (2026) based on [5].

The *Standard* provides the value orientation, the principles, the life cycle, and the Focus Areas; the *Guide* offers the detail on the domains, the processes, and the tailoring decisions. A third, dashed frame identifies the analytical grouping into four concentrations, which does not belong to PMI.

This separation eases reading but does not isolate concepts. The value delivery system affects choices in all domains; the processes of the *Guide* can be configured for different development approaches. Nor is there equivalence among principle, Focus Area, domain, and process: each term answers a distinct question.

2 STRUCTURE OF THE EDITION AND WHAT CHANGED

The 2025 edition brings together two volumes with different functions. The table summarizes what each part contains and where it is, with the page numbers of the printed edition [5].

Tabela 1 – Structure of PMBOK 8: Standard and Guide

Work and part	**Content**	**Where**
Standard — 1 Introduction	Purpose, characteristics of a project, organizational and project governance, operations and project management, relationship among portfolio, program, project, and operations	Ch. 1
Standard — 2 Value delivery system	Value creation, project environment, product management, functions and roles	Ch. 2
Standard — 3 Principles	Mindset and six principles, with impact, application, and connected domains	Ch. 3, p. 35

Work and part	Content	Where
Standard — 4 Life cycles	Phases, development approaches, selection criteria, delivery cadence, and five Focus Areas	Ch. 4, p. 57
Guide — 1 Introduction	Structure of the *Guide*, relationship with the *Standard*, and changes in the edition	Ch. 1
Guide — 2 Performance domains	Seven domains, each with key concepts, processes, tailoring, interactions, and verification of results	Ch. 2, p. 7
Guide — 3 Tailoring	What *tailoring* is, why and how to tailor, diagnostics	Ch. 3, p. 103
Guide — 4 Inputs and outputs	Catalog of example assets and artifacts	Ch. 4, p. 113
Guide — 5 Tools and techniques	Catalog of example tools and techniques	Ch. 5, p. 145

Source: Prepared by the authors (2026), based on [5].

The preface groups the changes in the edition into four categories: updating of terms and key concepts, refinement of the principles, reintroduction of the process groups as **Focus Areas**, and updating of the performance domains [5, Preface]. The edition presents six actionable principles, five Focus Areas, and seven domains that group 40 reformulated processes. The *Guide* synthesizes the former Knowledge Areas with other accepted concepts into seven domains and inserts tailoring considerations and examples in each one.

PMBOK 8 **does not mention the digital twin**: the term does not appear in the text of the edition. This matters for the use of this report in ICT projects, because the edition provides the management structure, not guidance specific to that technology. The literature on digital twins in project management is concentrated today in civil construction, as shown by the systematic and bibliometric review by Manzoor et al. [3], which analyzes critical success factors for their implementation.

3 VALUE DELIVERY SYSTEM, ENVIRONMENT, AND FUNCTIONS

Chapter 2 of the *Standard* describes where the project operates. **Value creation:** organizations create value for stakeholders, and the value expected from any project investment must equal or exceed a return threshold, financial or not, tangible or intangible. Business value is the quantifiable net benefit, in any tangible or intangible form, and value delivery today includes social impact and sustainability goals [5, Standard §2.1]. The assessment of a project's success considers two dimensions: the success of the project **outcomes**, that is, the realization of the intended value, and the success of the management **processes** [5, Standard §2.1.2].

Project environment. Two categories of influence act on the project: **enterprise environmental factors**, internal and external to the organization, and **organizational process assets**, which include processes, policies and procedures, and the knowledge repository. Among the external factors are legal constraints, such as laws and regulations on security, data protection, business conduct, employment, and procurement. The *Standard* also addresses organizational structures and their effects on the project manager and on resource availability, and product management considerations throughout the product life cycle [5, Standard §§2.2–2.3].

Functions and roles. People drive delivery by performing seven functions: providing oversight and coordination; requesting and managing *feedback*; facilitating and supporting; performing the work; applying specialized knowledge; providing organizational direction and vision; and supplying resources. They may be performed by one person, a team, or a combination of roles, and the form of coordination varies: decentralized, in self-organizing projects, or centralized, with a designated manager. The roles addressed are the project management team, the sponsor, the customer or product owner, the project team, and end users and other stakeholders [5, Standard §§2.4–2.5].

4 THE SIX PRINCIPLES AND THE MINDSET OF THE *STANDARD*

The *Standard* organizes a management mindset in three dimensions: **proactive**, **ownership**, and **value-oriented**. Each dimension integrates two principles [5, Standard §3.1].

Tabela 2 – Mindset dimensions and principles

Dimension	**Principles**
Proactive	**1AHV** *Adopt a Holistic View* and **3QPD** *Embed Quality Into Processes and Deliverables*
Ownership	**4BAL** *Be an Accountable Leader* and **6BEC** *Build an Empowered Culture*
Value-oriented	**2FOV** *Focus on Value* and **5SPA** *Integrate Sustainability Within All Project Areas*

Source: Prepared by the authors (2026), based on [5].

The table formulates, in working language, a useful question for applying each principle and a consequence for a decision; it does not replace PMI's definition [5, Standard §§3.3–3.8].

Tabela 3 – Practical questions and consequences of each principle

Official principle (code used in the course)	Practical question	Consequence for a decision
1AHV — Adopt a Holistic View	What relationships and effects beyond the immediate deliverable influence the result?	Examine dependencies among domains, organization, people, and environment.
2FOV — Focus on Value	What result justifies the work, and how do we know it remains valuable?	Compare alternatives by expected benefit, not only by the quantity of activities.
3QPD — Embed Quality Into Processes and Deliverables	How will quality be built and verified during the work?	Define criteria, reviews, and corrections early, avoiding inspection only at the end.
4BAL — Be an Accountable Leader	Who takes on and explains the difficult decisions?	Make responsibilities, criteria, and escalation visible.
5SPA — Integrate Sustainability Within All Project Areas	What effects persist for people, the environment, and the organization?	Include long-term consequences in choices about scope, resources, and risk.
6BEC — Build an Empowered Culture	Can the team act, learn, and raise problems?	Create trust, responsible autonomy, and feedback channels.

Source: Prepared by the authors (2026), based on [5].

The principles cut across the seven domains; there is no exclusive relationship between a principle and a single domain. The *Standard* states that all principles relate to all domains, and section 8 makes these interfaces explicit. The principles guide judgment: they are not additional processes.

5 LIFE CYCLES, DEVELOPMENT APPROACHES, AND CADENCE

Chapter 4 of the *Standard* addresses how work is organized over time. A project is decomposed into **phases**, which may be sequential, transitional, or overlapping. Each phase has attributes, such as name, duration, resource needs, and entry and exit criteria, and may end in a decision point (*phase gate*), at the end of the phase or, alternatively, at the beginning (*pre-phase gate*). The project life cycle is independent of the product life cycle. Tailoring the life cycle involves selecting the approach or mix of approaches, the types of processes and activities, and the attributes of a phase [5, Standard §4.1].

5.1 Predictive, adaptive, and hybrid

The **development approach** is the means used to create and evolve the product, service, or result over the life cycle. The three usual approaches form a spectrum, from predictive to adaptive, with hybrid as a combination of the two [5, Standard §4.2].

Tabela 4 – Development approaches: predictive, adaptive and hybrid

Approach	When it tends to serve	Characteristics
Predictive	Stable scope early on; large investments; regulatory oversight; cost of iterating greater than the value	Upfront planning, scope, schedule, and cost baselines; robust change control; may deliver incrementally
Adaptive	Requirements and technical solution with high uncertainty and volatility	Clear vision at the start; requirements progressively elaborated with *feedback*; periodic reviews in short cycles; some methods use iterations of 1 to 4 weeks
Hybrid	Uncertainty in only part of the scope; modularizable deliverables; several workstreams	Predictive and adaptive elements combined; the *Standard* describes four common patterns, such as adaptive development followed by predictive deployment or simultaneous use of both

Source: Prepared by the authors (2026), based on [5].

The *Standard* observes that, in practice, most projects can benefit from a hybrid approach and that focusing exclusively on one of the two fails to address part of the challenges of modern projects. This finding is consistent with the literature. A survey of 400 practitioners showed that few managers use agile methods without predictive techniques, that the sector and the type of project influence the choice, and that there is no absolute best approach [6]. In the analysis of data from the HELENA study, about two thirds of the projects surveyed combined agile and traditional methods to some degree [4]. A contingency-based approach identifies six ways of combining approaches according to suitability criteria and treats tailoring as the core of the topic [7].

5.2 How to choose and at what cadence to deliver

The choice of approach depends on variables of the **product**, the **project**, and the **organization**. The table summarizes the factors that the *Standard* lists [5, Standard §4.3].

Tabela 5 – Factors for choosing the development approach

Category	**Factors**
Deliverables	Degree of innovation; certainty of requirements; scope stability; ease of change; delivery options; risk; security requirements; *feedback*; regulation
Project	Stakeholders; schedule constraints; funding uncertainty; size and complexity; team experience and skills; interdependencies
Organization	Organizational structure; culture; organizational capability; team size and location

Source: Prepared by the authors (2026), based on [5].

The chosen approach conditions the **delivery cadence**: single delivery, multiple deliveries, periodic deliveries (common in adaptive approaches, on a regular calendar), or continuous delivery, in which tested and validated increments reach production on an ongoing basis [5, Standard §4.4]. The ISO 21502:2020 standard treats the subject in a similar way: it applies to any type of project, regardless of delivery approach and life cycle model, and moved from a process-based format to a practice-based format, replacing ISO 21500:2012 [2]. PMBOK 8 combines the two approaches: principles, Focus Areas, and domains, on one side, and 40 processes with example inputs and outputs, on the other.

6 FIVE FOCUS AREAS AND THE DISTRIBUTION OF THE PROCESSES

The *Project Management Focus Areas* are **Initiating (IN), Planning (PL), Executing (EX), Monitoring and Controlling (MC), and Closing (CL)**. They group management practices and are applicable regardless of application area, industry, and development approach. They do not impose a rigid sequence nor are they to be confused with life cycle phases. The processes, practices, or actions of each area are usually iterated before concluding a phase or the project [5, Standard §4.5].

Tabela 6 – The five Focus Areas

Area	**What it brings together**	**Guiding question**
IN Initiating	Actions to define a new project or phase, with formal authorization; initial scope, initial financial resources, stakeholders, and roles	What work must be authorized and who is involved?
PL Planning	Actions that establish the intended scope, refine objectives, and develop the course of action; repeated *feedback* loops	How do we integrate plans, requirements, resources, costs, communication, and risk responses?
EX Executing	Actions to complete the work according to the agreed course of action; this is where the choice of approach becomes most evident	How do we conduct the work, engage people, and implement responses?

Area	What it brings together	Guiding question
MC Monitoring and Controlling	Actions to track, measure, review, and regulate progress, identify necessary changes, and initiate them	How do we compare results with objectives and correct deviations?
CL Closing	Actions to formally conclude a phase, project, or contract, or to end it early; verification of outputs, results, and benefits	How do we close and transfer results to operations?

Source: Prepared by the authors (2026), based on [5].

The distribution of the 40 processes across the areas appears in Table 2-1 of the *Guide* and in the overview figures of each domain. *Identify Stakeholders* is the only Stakeholders process in Initiating, as shown by Figure 2-32 of the *Guide* [5, Guide §2.5.2]; with it, the totals by area add up to 2 + 19 + 8 + 10 + 1 = 40.

Tabela 7 – Process count by Focus Area

Code	Focus Area	Processes in PMBOK 8
IN	Initiating	2
PL	Planning	19
EX	Executing	8
MC	Monitoring and Controlling	10
CL	Closing	1
	Total	**40**

Source: Prepared by the authors (2026), from [5].

Figura 2 – 7 × 5 map of the 40 processes

Domain	IN	PL	EX	MC	CL	Total
Governance (GO)	1	2	3	2	1	9
Scope (SO)	0	4	0	2	0	6
Schedule (SH)	0	2	0	1	0	3
Finance (FI)	0	3	0	1	0	4
Stakeholders (ST)	1	2	2	2	0	7
Resources (RE)	0	2	2	1	0	5
Risk (RI)	0	4	1	1	0	6
Total	**2**	**19**	**8**	**10**	**1**	**40**

Source: [5, Guide, Table 2-1 and domain overview figures]; own calculation and presentation.

The cells are **counts**, not intensity of importance.

Planning concentrates 19 processes, but this number does not mean that it takes up almost half of a project's time. The table classifies processes, not duration, cost, or

relevance. Governance appears in all five areas, whereas the other domains have more concentrated distributions. Control may update decisions in any domain; a row with zero in a given area does not mean that the topic disappears from it.

7 THE SEVEN PERFORMANCE DOMAINS OF THE *GUIDE*

A performance domain is a group of related processes that are essential for the effective delivery of value. The seven domains are interactive, interrelated, and interdependent, and operate as an integrated system, **without specific order or weighting** [5, Guide §2]. For each one, the *Guide* presents key concepts, processes, tailoring considerations, interactions with other domains, and verification of results (Tables 2-5 to 2-11 of the *Guide*). They are not isolated departments.

Governance (GO, 9 processes) is the structure, functions, and processes that guide management decisions and activities to optimize value delivery. It is holistic and integrating, considers all the other domains, includes elements of integration management (strategic alignment, decision and change, success criteria), and **integrates risk and opportunity management**. It also ensures the alignment of the project with the organization's portfolio and strategy [5, Guide §2.1].

Scope (SO, 6) ensures that the project contains all the work required, and only that work, to maximize value. The domain **includes quality**: regarding deliverables, it addresses acceptance criteria and the meeting of stakeholder expectations and of project and product requirements [5, Guide §2.2; Standard §3.2].

Schedule (SH, 3) establishes a plan for how and when the project will deliver products, services, and results. It serves as a tool for communication, expectation management, and the basis for performance reports, and it facilitates early identification of delays and risks [5, Guide §2.3].

Finance (FI, 4) covers costs, funding, and, in some cases, the project's value proposition: planning, estimating, budgeting, funding, managing, measuring, and controlling costs to optimize value for the organization [5, Guide §2.4].

Stakeholders (ST, 7) addresses engagement, from the identification of stakeholders to the monitoring of engagement throughout the life cycle, and is linked to communications management. Stakeholders include the sponsor, the team, and all people affected, or who feel they may be affected, by the project; they can number from a few to millions [5, Guide §2.5].

Resources (RE, 5) refers to how the team plans and uses the available resources: people, equipment, materials, facilities, infrastructure, software, test environments, licenses, services, and information. It also covers availability, utilization, and maintenance [5, Guide §2.6].

Risk (RI, 6) is a comprehensive approach to creating project resilience, with

the capacity to anticipate, prepare for, respond to, and adapt to risks and disruptions, proactively and with flexible responses [5, Guide §2.7].

7.1 Auditable inventory of the 40 processes

The table keeps the **official English names**. The Portuguese text in parentheses serves reading and is not a translation authorized by PMI. The **concentration** column carries the analytical grouping (section 10), which does **not** belong to PMI. The presence of a process in the inventory does not oblige its formal execution in every project; its application depends on *tailoring* [5, Guide Table 2-1 and §§2.1–2.7].

Tabela 8 – Inventory of the 40 PMBOK 8 processes

#	Domain	Area	Concentration	Official process — Working translation (Portuguese)
1	GO	IN	Consolidation	*Initiate Project or Phase* (iniciar projeto ou fase)
2	GO	PL	Consolidation	*Integrate and Align Project Plans* (integrar e alinhar planos do projeto)
3	GO	PL	Consolidation	*Plan Sourcing Strategy* (planejar estratégia de suprimento)
4	GO	EX	Consolidation	*Manage Project Execution* (gerenciar execução do projeto)
5	GO	EX	Consolidation	*Manage Quality Assurance* (gerenciar garantia da qualidade)
6	GO	EX	Consolidation	*Manage Project Knowledge* (gerenciar conhecimento do projeto)
7	GO	MC	Consolidation	*Monitor and Control Project Performance* (monitorar e controlar desempenho)
8	GO	MC	Consolidation	*Assess and Implement Changes* (avaliar e implementar mudanças)
9	GO	CL	Consolidation	*Close Project or Phase* (encerrar projeto ou fase)
10	SO	PL	Birth	*Plan Scope Management* (planejar gestão do escopo)
11	SO	PL	Birth	*Elicit and Analyze Requirements* (elicitar e analisar requisitos)
12	SO	PL	Birth	*Define Scope* (definir escopo)
13	SO	PL	Birth	*Develop Scope Structure* (desenvolver estrutura do escopo)
14	SO	MC	Birth	*Monitor and Control Scope* (monitorar e controlar escopo)
15	SO	MC	Birth	*Validate Scope* (validar escopo)
16	SH	PL	Birth	*Plan Schedule Management* (planejar gestão do cronograma)
17	SH	PL	Birth	*Develop Schedule* (desenvolver cronograma)
18	SH	MC	Birth	*Monitor and Control Schedule* (monitorar e controlar cronograma)
19	FI	PL	Means	*Plan Financial Management* (planejar gestão financeira)

#	Domain	Area	Concentration	Official process — Working translation (Portuguese)
20	FI	PL	Means	*Estimate Costs* (estimar custos)
21	FI	PL	Means	*Develop Budget* (desenvolver orçamento)
22	FI	MC	Means	*Monitor and Control Finances* (monitorar e controlar finanças)
23	ST	IN	Who	*Identify Stakeholders* (identificar partes interessadas)
24	ST	PL	Who	*Plan Stakeholder Engagement* (planejar engajamento das partes interessadas)
25	ST	PL	Who	*Plan Communications Management* (planejar gestão das comunicações)
26	ST	EX	Who	*Manage Stakeholder Engagement* (gerenciar engajamento das partes interessadas)
27	ST	EX	Who	*Manage Communications* (gerenciar comunicações)
28	ST	MC	Who	*Monitor Stakeholder Engagement* (monitorar engajamento das partes interessadas)
29	ST	MC	Who	*Monitor Communications* (monitorar comunicações)
30	RE	PL	Means	*Plan Resource Management* (planejar gestão dos recursos)
31	RE	PL	Means	*Estimate Resources* (estimar recursos)
32	RE	EX	Means	*Acquire Resources* (adquirir recursos)
33	RE	EX	Means	*Lead the Team* (liderar a equipe)
34	RE	MC	Means	*Monitor and Control Resourcing* (monitorar e controlar disponibilização de recursos)
35	RI	PL	Who	*Plan Risk Management* (planejar gestão dos riscos)
36	RI	PL	Who	*Identify Risks* (identificar riscos)
37	RI	PL	Who	*Perform Risk Analysis* (realizar análise de riscos)
38	RI	PL	Who	*Plan Risk Responses* (planejar respostas a riscos)
39	RI	EX	Who	*Implement Risk Responses* (implementar respostas a riscos)
40	RI	MC	Who	*Monitor Risks* (monitorar riscos)

Source: Prepared by the authors (2026), based on [5]; the concentration is the analytical grouping.

The inventory reveals concrete dependencies. A Scope requirement may demand resources, budget, and a Governance decision; schedule changes may alter risks and stakeholder expectations. The fact that a process sits under a domain and a Focus Area makes it easier to locate, but does not limit its interactions.

7.2 Tailoring considerations by domain

Each domain carries a section of tailoring considerations (section 2.x.3 of the *Guide*, except Governance, which is 2.1.7). The table summarizes one characteristic consideration for each; the complete lists are in the book [5, Guide §§2.1.7, 2.2.3, 2.3.3, 2.4.3, 2.5.3, 2.6.3, and 2.7.3].

Tabela 9 – Characteristic tailoring consideration of each domain

Domain	**Characteristic tailoring consideration**
Governance	Tailor to the industry, regulatory considerations, the organization, and the project context; identify committees and status reporting requirements
Scope	Dependence on external partners (how to integrate scope into contractual agreements) and the dynamism of the environment, which calls for flexibility in scope management
Schedule	Chosen life cycle and approach: in predictive, schedule defined early and baseline approved; attributes of the product, the team, and the culture
Finance	A product in a regulated industry may require formal controls, such as compliance with SOX or GDPR; an iterative approach flattens the spending curve
Stakeholders	Organizational culture is the main factor: flat organizations use open and informal channels; hierarchical ones, more formal approaches
Resources	Leadership styles according to the project; in predictive, resources planned early; in adaptive, self-organizing teams
Risk	Size and complexity determine whether a detailed approach is needed or a simplified process suffices

Source: Prepared by the authors (2026), based on [5].

8 HOW THE PRINCIPLES CUT ACROSS THE DOMAINS

The *Connected Performance Domains* subsections of the *Standard* (§§3.3.3 to 3.8.3) describe, for each principle, how it connects to each of the seven domains. In each of the six sections the text addresses the seven domains, which gives **42 expressed connections**. The *Standard* adds that the holistic view is supported by all domains; that quality and focus on value are addressed mainly in Governance, Scope, Risk, Schedule, Finance, and Stakeholders; and that the accountable leader cuts across Governance, Stakeholders, and Risk [5, Standard §3.2].

The matrix in Figure 3 summarizes **one characteristic contribution per cell**. It is an analytical paraphrase of PMBOK 8 passages, not a scale of strength, priority, or compliance. **Legend:** all cells have an express basis in the text of the *Standard*; the short wording is the group's.

Figura 3 – Matrix of principle relationships

Principle	GO	SO	SH	FI	ST	RE	RI
1AHV Holistic view	transparent, data-based decision-making	aligns activities to scope	considers interdependencies	aligns budget to priorities	inclusive engagement	integrates components to maximize value	risks across all domains and their interdependencies
2FOV Value	governance sized to value, without bureaucracy	all work contributes to value; avoids excess	timeline linked to expected value	financial resources to maximize value	value from the stakeholders' perspective	decisions guided by the value proposition	pragmatic and effective responses
3QPD Quality	structures that ensure quality standards	especially strong connection with deliverables	quality reviews in the schedule	avoids rework and waste	feedback integrated into decisions	the right resources maintain thresholds	quality risks anticipated
4BAL Accountable leadership	transparent and ethical decisions	resolve disagreements about including or excluding	discussions with the right parties	financial pressures handled responsibly	obtain and maintain support	manage with responsibility and integrity	act with integrity in the face of threats and opportunities
5SPA Sustainability	sustainability goals in governance	sustainable requirements in scope	decisions consider environmental impact	long-term costs and benefits	education and collaboration on sustainability	access to resources aligned to requirements	environmental and social threats identified and mitigated
6BEC Empowered culture	transparent channel between team and governance	adjust requirements with open communication	team suggests speeding up, slowing down, or stopping	less unplanned spending, focus on value	team defines the level of engagement	access to resources according to requirements	team defines risk limits and participates

Bands: blue = proactive dimension; turquoise = ownership dimension; gray = value-driven dimension.

Source: Prepared by the authors (2026) based on [5, Standard §§3.2 to 3.8].

In the figure, background color distinguishes the three pairs of the *Standard*'s mindset, and the short text keeps the reading accessible without color. The matrix does not create a numerical performance index.

This reading avoids two errors. The first would be to turn principles into additional processes. The second would be to take the domains as silos: a quality choice in Scope may require budget, skills, and a risk response.

9 TAILORING

Chapter 3 of the *Guide* defines *tailoring* as the deliberate adaptation of the management approach, governance, and processes to align them with the project's environment and objectives. Tailoring considers the development approach, processes, life cycle, deliverables, and the way stakeholders and the team are involved. It is guided by the principles of the *Standard*: organizations with low risk tolerance tend to adopt structured processes

and oversight mechanisms, whereas those with higher tolerance may prioritize agility and decision-making autonomy [5, Guide §3.1].

9.1 Why tailor

There is no single way to manage all projects. Too few processes may leave critical decisions unaddressed; too many processes are costly and waste effort. Tailoring brings three benefits noted by the *Guide*: greater commitment from team members who helped tailor the approach; customer focus, because their needs influence the build; and more efficient use of project resources. Criticality, scale, duration, complexity, industry standards, culture, number of stakeholders, and organizational maturity influence the choice. There are situations in which tailoring is limited, for example when organizational policy or a contract imposes an approach [5, Guide §§3.1–3.2].

9.2 What to tailor

Three objects can be tailored [5, Guide §3.3]:

- **Life cycle and development approach.** Choose phases, cadence, and a predictive, adaptive, or hybrid combination. A project may have, for example, a predictive approach for constructing a building and an adaptive one for defining the computing capabilities; from the project's point of view, this is a hybrid approach.
- **Processes.** Five operations can be applied to the elements of a process: **add**, to address specific conditions or increase rigor; **modify**, to adjust inputs, outputs, and tools; **remove**, to reduce unnecessary cost or effort; **combine**, to add value; and **align**, to ensure consistency of definition and application.
- **Engagement.** People (assess the skills of leadership and the team), empowerment (which responsibilities and local decisions stay with the team), and integration (how to form a team from diverse contributors, internal and external).

These operations do not authorize omitting a legal or contractual requirement; they help configure management practice under real constraints.

Hypothetical examples of the five operations. The table illustrates, in an ICT project with a digital twin, how the operations could be used. They are working examples, **not PMBOK 8 prescriptions**.

Tabela 10 – Hypothetical examples of the five operations on processes

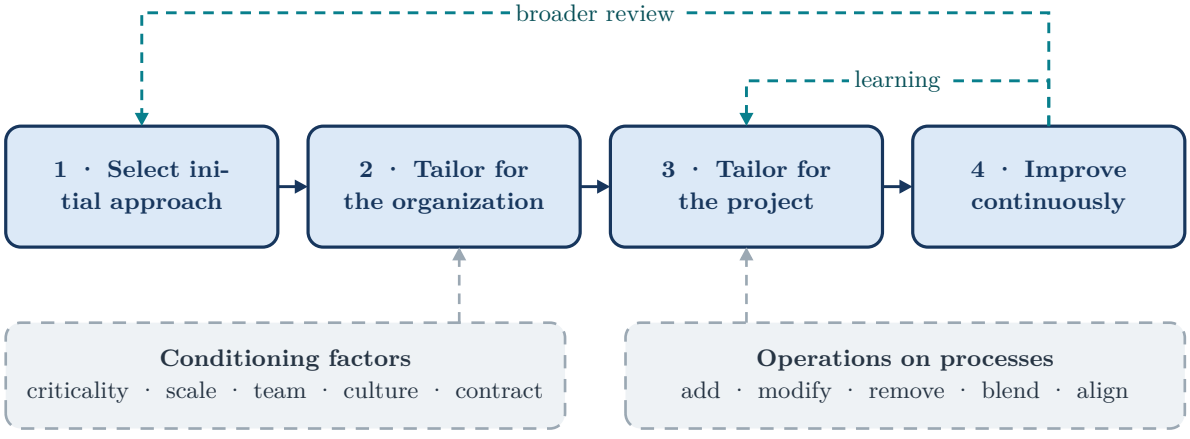
Operation	Process	Hypothetical example
Add	*Identify Risks*	Add risk categories for data and for integration between information and operations systems
Modify	*Plan Communications Management*	Adjust channels and frequency for field operators with limited connectivity
Remove	*Plan Sourcing Strategy*	Waive formalization if there is no significant external procurement, keeping the applicable contractual requirements
Combine	*Plan Scope Management* and *Plan Schedule Management*	Merge them into one integrated plan, in a small project, without losing the traceability of each
Align	*Manage Communications*	Unify terms and records across partner organizations

Source: Prepared by the authors (2026); illustrative examples, not PMBOK 8 prescriptions.

9.3 The tailoring process in four steps

The process begins with an analysis of the environment and follows four steps, with feedback [5, Guide §3.4]: (1) **choose the initial development approach**; (2) **tailor for the organization**; (3) **tailor for the project**; and (4) **implement continuous improvement**, with inspection and adaptation.

Figura 4 – Tailoring cycle



Source: Prepared by the authors (2026) based on [5, Guide §3.4 and Figure 3-1].

The path begins with the choice of development approach, passes through adjustment to the organization and to the project, and returns to earlier decisions through continuous improvement. Constraints such as criticality, scale, team, culture, and contract influence the adjustment; the five operations on processes apply to the adjustment to the project.

Step 1. The choice considers the product, the delivery cadence, and awareness of the available options. A **suitability filter** is a decision tool, not a rigid method: it assesses criteria of culture, team dynamics, and project factors to discuss the initial approach among predictive, adaptive, and hybrid. Silvius-Zuchi and Silvius propose suitability criteria for designing hybrid approaches [7].

Step 2. Organizations may offer methodologies and approaches as a starting point, and those with process governance must ensure that tailoring is aligned with policy. The project management office (PMO), where one exists, has a role in reviewing and approving tailored approaches: tailoring that affects only the team requires less oversight than tailoring that affects external groups. In projects with several organizations, tailoring must use a multi-organizational approach, harmonizing the processes, governance, and approaches of each participant.

Step 3. Three groups of attributes guide the adjustment to the project, each with guiding questions: the **product or deliverable** (eight attributes, among them compliance with standards, product type, industry market, technology, timeline, stability of requirements, security, and possibility of incremental delivery), the **team** (six, among them size, geography, organizational distribution, experience, customer access, and diversity), and the **culture** (buy-in, trust, empowerment, and organizational culture).

Step 4. Results, retrospectives, and other information maintain or revise the choices; the *Guide*'s figure uses the labels "inspect and adapt" and "improvement".

9.4 Tailoring the domains and running diagnostics

The work of each domain can also be tailored according to the project's uniqueness; the principles guide practitioners' behavior in doing so, and examples for each domain are in section 2 of the *Guide* [5, Guide §3.5; Figure 3-4]. **Diagnostics** use retrospectives or lessons learned to assess the effectiveness of current approaches; teams that do not use retrospectives can turn to issues, threats, quality metrics, and stakeholder *feedback*. Table 3-1 of the *Guide* relates eight common situations to tailoring suggestions [5, Guide §3.6]. The table summarizes the triggers and the suggested response.

Tabela 11 – Common situations and tailoring suggestions

Situation	**Tailoring suggestion**
Low-quality deliverables	Analyze causes and apply targeted *feedback* and quality assurance measures, focused on improving processes and not only on increasing checks
Team unsure about how to work	Assess the team's needs and offer guidance, mentoring, training, and knowledge sharing
Team uncooperative and in silos	Integration activities and one-on-one conversations; strengthen bonds and coordination

Situation	Tailoring suggestion
Long waits for approvals	Simplify approval, designating people authorized to decide up to certain value limits
Excess work in progress or waste	Value stream mapping and *kanban* boards to visualize the work and find solutions
Stakeholders poorly engaged or giving negative *feedback*	Check whether they receive timely and relevant information and tailor the communication methods
Lack of transparency about progress	Check that the right data are collected, analyzed, shared, and discussed, and validate the measures with the team and stakeholders
Problems and risks for which the team is unprepared keep arising	Investigate the causes to identify gaps in processes or activities

Source: Prepared by the authors (2026), based on [5].

In summary, chapter 3 of the *Guide* describes tailoring as the analysis and modification of people, processes, and tools to the project's environment and objectives [5, Guide §3.7].

10 FOUR CONCENTRATIONS AS AN ANALYTICAL GROUPING

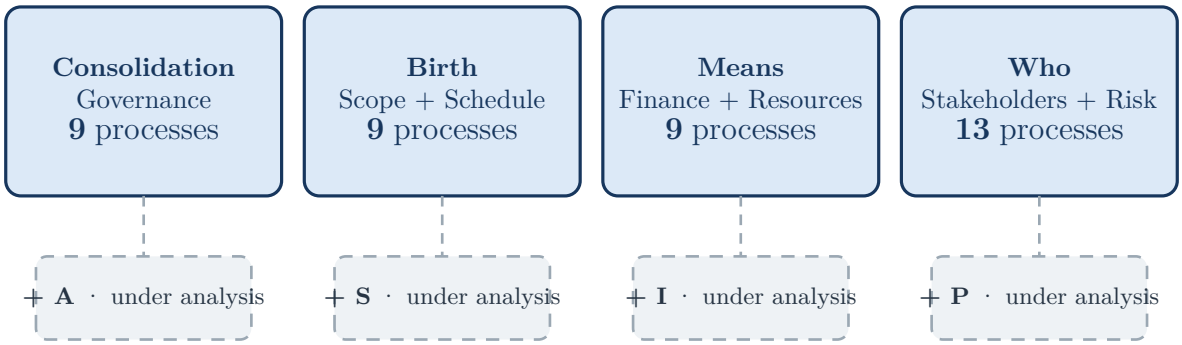
To present the seven domains in blocks, this report uses the **four analytical concentrations adopted in the GPTI/PPCA course**: **Consolidation** brings together Governance; **Birth** brings together Scope and Schedule; **Means** brings together Finance and Resources; **Who** brings together Stakeholders and Risk. This grouping is **not an official component of PMBOK 8**: it corresponds to the didactic organization of the course, used here as a reading reference. The labels A, S, I, and P indicate, respectively, proposed extensions in business analysis, sustainability, innovation, and privacy under analysis; they are not included in the sum of the 40 processes nor are they described here as new PMI processes.

Tabela 12 – Analytical concentrations and process count

Concentration	Domains	Processes	Extension under analysis (outside PMI)
Consolidation	Governance	9	A
Birth	Scope and Schedule	9	S
Means	Finance and Resources	9	I
Who	Stakeholders and Risk	13	P
Total	Seven domains	**40**	

Source: Prepared by the authors (2026).

Figura 5 – Grouping of the seven domains into four concentrations



Source: Domains and counts: [5, Guide, Table 2-1]. Grouping and labels: didactic organization of the GPTI/PPCA course.

The numbers 9, 9, 9, and 13 add up only existing processes. Cards with dashed outlines mark extension themes and prevent them from being confused with official domains.

The grouping facilitates conversation and the distribution of work, but it does not change the way the *Guide* presents the domains. Governance continues to interact with all the others; a decision about people may involve scope, time, finance, and resources. So that the analytical map does not become an artificial barrier, the connections among the seven domains should be read together with the tables of processes and principles.

11 WHERE PMBOK 8 ALREADY TOUCHES PERSONAL DATA AND PRIVACY

Since the report will also be read by those who work with projects that process personal data, this table records, **without proposing a new process**, the points at which the edition already mentions the topic. The references are working paraphrases [5].

Tabela 13 – PMBOK 8 points that touch personal data

Where	What the edition says	Section
Enterprise environmental factors	Legal constraints include laws and regulations on security, data protection, business conduct, employment, and procurement	*Standard* §2.2.1
Product attributes in tailoring	The “security” attribute asks whether elements of the product are confidential or classified	*Guide* §3.4.3.1
Finance tailoring	Regulated industries may require formal compliance controls, such as SOX or GDPR [1], for the integrity and security of financial data	*Guide* §2.4.3

Where	What the edition says	Section
Manage communications	The process covers collection, creation, distribution, storage, retrieval, monitoring, and final disposition of project information	*Guide* §2.5.2
AI in project management	AI systems use large datasets that may be sensitive and regulated by privacy policies and laws; data must be protected and collected under a privacy policy	*Guide*, Annex X3.3

Source: Prepared by the authors (2026), based on [5].

The edition does **not** define data subject, legal basis, privacy impact assessment, or deletion of personal data as management objects; the term "digital twin" does not appear either. This record delimits what the synthesis claims: PMBOK 8 offers the management structure and mentions the topic at scattered points, and a proposal for an organization specific to privacy would be a work distinct from this one.

12 CONCLUSION

The eighth edition combines **principle-based guidance** with **technical detail** and insists on tailoring. This combination avoids two extremes: using a list of processes as a universal recipe or invoking principles without explaining how decisions will be tracked. The *Standard* offers the value delivery system, the six principles, the life cycle, the development approaches, and the five Focus Areas; the *Guide* distributes 40 processes across seven domains and describes how to tailor them. The table of 40 processes shows possibilities for action; the tailoring cycle explains why their form, intensity, and combination vary. The literature consulted converges with this view: hybrid approaches are frequent, the choice depends on context, and tailoring is the core of the topic [4, 6, 7].

The four concentrations adopted in the GPTI/PPCA course are a way of reading, not a replacement for PMI's official index. The tailoring examples are hypothetical and do not demonstrate realized benefit. This synthesis likewise neither validates an extension nor measures project performance. A concrete application would need to choose indicators, responsibilities, artifacts, and review frequency, in addition to verifying the organization's and the contract's own constraints. Since PMBOK 8 does not address digital twins, a project of that nature must complement the management structure with specific technical references.

12.1 Use of artificial intelligence

This work used generative AI (ChatGPT Astral 6 Sol and Sonnet 5.5, both in high mode) to support text editing, formatting, figure creation, and table drafting. We follow the COPE position on authorship and AI tools (COPE, 2023), which asks authors to disclose

the use and to remain responsible for the work. Three points guide how to read this statement.

AI is neither an author nor a source. No claim about PMBOK 8, the LGPD, or the literature should be accepted because a language model produced it. Each claim points to its source (section, table, regulation, or reference) so that readers can check it.

Known risks of this use. Language models can invent references, swap numbers, and soften caveats, and translation between Portuguese and English can shift technical meanings. References, counts, and process names are therefore the points that most deserve checking against the sources, and this report does not replace reading the PMBOK 8 edition.

Responsibility. The group is responsible for the content, including the parts produced with AI support. The statement describes categories of use, does not log each interaction, and the team's effort was not measured.

Relation to the topic. PMBOK 8 does not organize the use of AI as a domain, but it devotes an appendix (X3) to it [5]. A project that adopts AI should treat it like any other element to tailor, not as an exception to the method.

COMMITTEE ON PUBLICATION ETHICS (COPE). *Authorship and AI tools: COPE position statement*. 13 Feb. 2023. Available at: `https://publicationethics.org/guidance/cope-position/authorship-and-ai-tools`.

13 FINAL REPORT OF THE WORK

Following the format of the *Final report*, an output of the *Close Project or Phase* process [5, Guide §2.1.6.9], this item summarizes the performance of the work of preparing this synthesis, taken as the project reported.

Tabela 14 – Final report of the work

Final report item	**Status of the work**
Summary description	Preparation of a synthesis of the PMBOK 8 *Standard* and *Guide*: structure of the edition, value delivery system, six principles, life cycles and approaches, five Focus Areas, seven domains and 40 processes, tailoring, analytical grouping into concentrations, and points of contact with personal data.
Scope objectives and evidence of completion	Scope: cover the PMBOK 8 elements above, with an emphasis on *tailoring*. Completion criterion: sections 2 to 11 produced. Evidence: master table of the 40 processes, 7 × 5 map, 6 × 7 matrix, tailoring cycle, and diagnostics table. Limit: the report does not reproduce chapters 4 and 5 of the *Guide*, which are reference catalogs.

Final report item	Status of the work
Quality objectives and verification	The document presents the names and counts of the 40 processes, with reference to sections 2.1 to 2.7, Table 2-1, and the domain overview figures of the *Guide*, as well as the 42 principle-domain connections associated with sections 3.3.3 to 3.8.3 of the *Standard*. The final check of these correspondences and of the equivalence between the Portuguese and English versions must be confirmed by the authors. There is no record of external review.
Cost objectives and variances	The work had no financial budget. The team's effort is not reported here.
Validation of the final result	Documentary synthesis produced; final collective review and confirmation of the documentary checks still to be recorded. Validation by project management experts still pending.
Schedule and realization of benefits	Work carried out in a short cycle of preparation, review, and checking, with delivery scheduled for 6 November 2026. The benefits of applying the synthesis are hypotheses to be tested.
Business needs met	The need for an adaptable PMBOK 8 synthesis as a basis for project management guides received a structured response; its effectiveness depends on validation.
Risks and issues encountered	(i) Confusion between official structure and analytical grouping: addressed with separate labels and figures. (ii) Translation and oversimplification: addressed with official English names and caveats. (iii) Restricted-access source: addressed by citing sections, tables, and figures. (iv) Risk of keeping claims or references without final verification: mitigation proposed by checking the sources, the counts, and the PT/EN equivalence before submission.

Source: Prepared by the authors (2026), following the *Final report* format of [5].

REFERENCES

[1] EUROPEAN UNION. **Regulation (EU) 2016/679 of the European Parliament and of the Council, of 27 April 2016** (General Data Protection Regulation). **Official Journal of the European Union**, L 119, 4 May 2016.

[2] ISO — INTERNATIONAL ORGANIZATION FOR STANDARDIZATION. **ISO 21502:2020: Project, programme and portfolio management — Guidance on project management**. Geneva: ISO, 2020.

[3] MANZOOR, B.; ANTWI-AFARI, M. F.; ALOTAIBI, K. S.; EDWARDS, D. J.; POSILLICO, J. J. Critical success factors for implementing digital twin technology in construction projects: a systematic and bibliometric approach. **Journal of Construction Engineering and Management**, vol. 151, art. 031250, 2025. DOI: 10.1061/JCEMD4.COENG-17248.

[4] NOLL, J.; BEECHAM, S. How agile is hybrid agile? An analysis of the HELENA data. In: FRANCH, X.; MÄNNISTÖ, T.; MARTÍNEZ-FERNÁNDEZ, S. (ed.). **Product-Focused Software Process Improvement: 20th International Conference, PROFES 2019**. Cham: Springer, 2019. pp. 341-349. (Lecture Notes in Computer Science, vol. 11915). DOI: 10.1007/978-3-030-35333-9_25.

[5] PMI — PROJECT MANAGEMENT INSTITUTE. **The Standard for Project Management and A Guide to the Project Management Body of Knowledge (PMBOK Guide)**. 8th ed. Newtown Square: PMI, 2025. Restricted-access document, obtained by purchase or PMI membership.

[6] REED, A.; ANGOLIA, M.; BAHAM, C.; IGAH, R. Usage of hybrid project management approaches and influences on approach selection. **Communications of the Association for Information Systems**, vol. 54, pp. 402-427, 2024. DOI: 10.17705/1CAIS.05415.

[7] SILVIUS-ZUCHI, D.; SILVIUS, G. Designing a fitting hybrid project management approach: a contingency perspective. **International Journal of Information Systems and Project Management**, vol. 14, no. 1, art. 4, 2026.